

Go Mountain! VR: Virtual Poles and Physical Motions for Trekking and Skiing Experiences

Yukiya Ojima
ojima-yukiya@ed.tmu.ac.jp
Tokyo Metropolitan University
Hino-shi, Tokyo, Japan

Vibol Yem
yem@iit.tsukuba.ac.jp
University of Tsukuba
Tsukuba-shi, Ibaraki, Japan

Shogo Okamoto
okamotos@tmu.ac.jp
Tokyo Metropolitan University
Hino-shi, Tokyo, Japan

Yasushi Ikei
ikei@vr.u-tokyo.ac.jp
The University of Tokyo
Bunkyo-ku, Tokyo, Japan

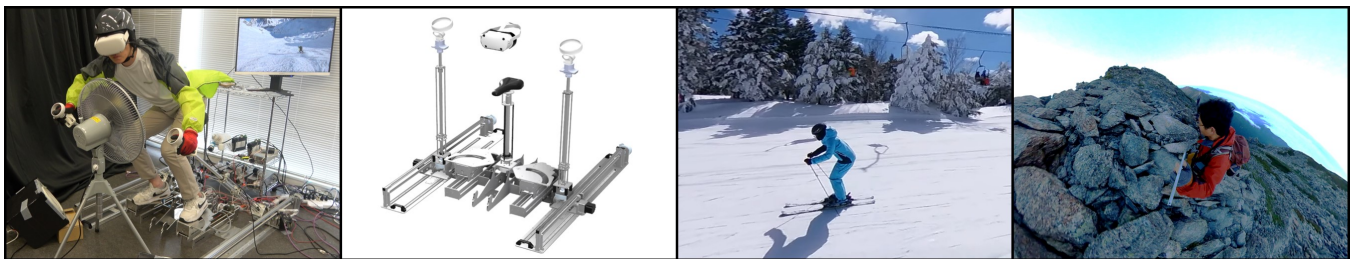


Figure 1: Go Mountain! VR experience simulator (L) with 360-degree movies of skiing and mountain trekking (R).

Abstract

A novel immersive experience system that creates the physical sensation of trekking and skiing within a 360 video is presented. This system incorporates full-body motion, including the use of handheld poles, through various forms of stimulation: vestibular, lower/upper limbs, airflow, and vibration at the feet, saddle, and hands. These sensations are synchronized with the audiovisual scenery relayed through the head-mounted display, offering conference participants an unique experience of mountain activities.

CCS Concepts

• Human-centered computing → Virtual reality; Haptic devices.

Keywords

Full body VR, Passive experience, Motion poles, Trekking, Skiing

ACM Reference Format:

Yukiya Ojima, Shogo Okamoto, Vibol Yem, and Yasushi Ikei. 2024. Go Mountain! VR: Virtual Poles and Physical Motions for Trekking and Skiing Experiences. In *SIGGRAPH Asia 2024 Emerging Technologies (SA Emerging Technologies '24)*, December 03–06, 2024, Tokyo, Japan. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3681755.3688940>

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA Emerging Technologies '24, December 03–06, 2024, Tokyo, Japan

© 2024 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1137-4/24/12

<https://doi.org/10.1145/3681755.3688940>

1 Introduction

Virtual Reality (VR) offers a diverse array of experiences, among which mountain activities are notably engaging due to their dynamic physical requirements and visually stimulating environments. These experiences are well suited for multisensory simulations that go beyond mere video representation, as they incorporate bodily sensations, which significantly enhance the sense of presence and offer practical training benefits.

Ski simulators, for example, have been developed for skill acquisition [Matsumoto et al. 2022; Wu et al. 2019] and entertainment through various research and gaming applications. In training contexts, users engage with the virtual environment through active physical movements. Conversely, in gaming, the replication of bodily sensations is not always necessary, though self-navigation is a typical feature.

In previous work, we introduced a system that recreated bodily experiences for users by passively delivering synchronized audiovisual and physical stimuli [Shimizu et al. 2018]. Grounded in the principles of passive presentation, we build an entirely novel system, as illustrated in Figure 1, that integrates hand movements for immersive, first-person experiences of mountain activities. One of an enhanced feature of this system is the precise stimulation of pole handling motion.

2 Multisensory VR System

Figure 2 shows the device we designed, which generates a sense of physical movement with 11 degrees of freedom (vibrators omitted). Our newly designed device creates a comprehensive sensory experience of mountain trekking and skiing, not just through visual engagement but by stimulating the entire body. It simulates

scenarios such as climbing Kitadake—Japan’s second-highest mountain—and skiing in Shiga Kogen (Plateau). The system can also adapt to a variety of environmental conditions provided in the video.

The setup includes a head-mounted display (HMD) for audiovisual input, extendable and tilting poles for simulating hand movements, a saddle to generate vertical acceleration for the torso, and sliders and pedals that replicate lower limb posture and motion. Additionally, fans produce airflow relevant to the skiing speed or mountainous weather conditions, and vibrators are attached to the pedals, saddle, and hand controllers to enhance tactile feedback.

These components work in coordination to simulate the physical sensations associated with trekking or skiing. The HMD enhances the realism by displaying stereoscopic videos for specific scenes, effectively immersing participants in the virtual environment.

Figure 3 illustrates the preliminary trajectory of each degree of freedom during the skating and skiing simulations. The up-and-down movement of the saddle, the back-and-forth motion of the leg sliders, and the reciprocating rotation of the pedal pitch create a sense of movement for the torso and legs, featuring different phases and waveforms. The poles’ extension and contraction include moments of sudden stops, mimicking the sensation of the tip touching the ground, followed by rapid tilting and backward movement to simulate propulsion by the poles.

When shoes touch the ground, step on snow, the contact state between skis and the snow surface changes, or the pole tips touch the ground, vibrational tactile feedback is conveyed to the saddle, soles, and hand controllers, enhancing the spatial events.

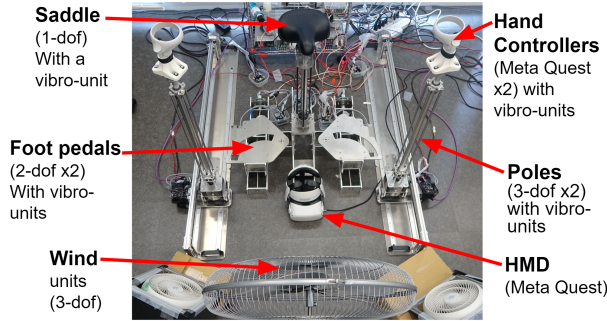


Figure 2: “Go Mountain! VR” experience simulator.

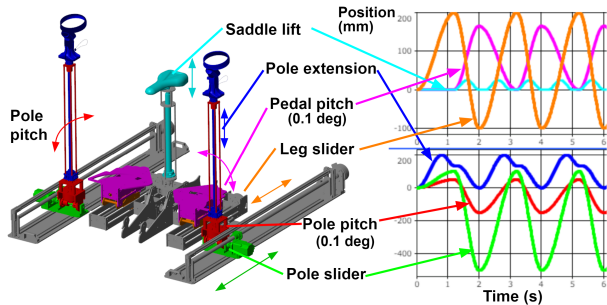


Figure 3: Structure model and typical trajectories of translations and rotations. (before optimization)



Figure 4: First-person view of a slope scene.

3 Experience at the Venue

This system enables participants to experience the sensation of movement involving the entire body, including the stabilization of the upper body and the propulsion of the hands during mountain walking and skiing. The motion and vibrational stimuli from the saddle and lower limbs, along with the movement of the poles held by the hands, contribute to this immersive experience.

In mountain walking, the saddle and feet move with relatively slow cycles and large amplitudes, enabling a pronounced sensation of the ground through the poles. The poles’ extension and contraction replicate changes in the ground height where the tip lands, with the landing position and angle of the poles adjustable by the user. The sensation of ascent is enhanced by the vestibular sensation from the saddle, the leg movements, and the vibration upon landing.

For skiing movements, the current system configuration is most suited to a form resembling cross-country skiing, which is closer to walking. Users alternate bringing their feet forward in a skating motion or keep their feet together and use the poles for propulsion. Downhill skiing is also possible by keeping the feet together.

Figure 4 depicts a scene from an omnidirectional 4K video in a subjective view. Such immersive videos inducevection and promote an integrated perception through physical motion. In all scenarios, the handgrip of the poles provides sensations and vibrations from the pole work and ground contact, complemented by the walking sensations and landing feel for the entire body, along with the airflow sensations of the ski resort or mountains. These enhance the realism of movement within the immersive video space.

Participants at the conference will have the option to choose between experiences with slow viewpoint movement for minimalvection and those with somewhat faster viewpoint movements. Walking in gently sloped mountain areas or cross-country skiing on gentle slopes can offer experiences with significantly reduced occurrence of motion sickness. Alternatively, participants can opt for steep mountain walking (Kitadake) and fast skiing experiences, both involving significant viewpoint movement. Each experience lasts about 3 min to prevent long waiting times. We look forward to receiving comments from participants on-site. This system poses no safety issues if used correctly, but comprehensive safety measures will be in place.

Acknowledgments

This work was partially supported by JSPS KAKENHI JP19K20325, 21H04883, 18H04118, 21K19785 in addition to Local-5G Project at TMU and the Telecommunications Advancement Foundation.

References

- Takashi Matsumoto, Erwin Wu, and Hideki Koike. 2022. Skiing, Fast and Slow: Evaluation of Time Distortion for VR Ski Training. In *Proceedings of the Augmented Humans International Conference 2022* (Kashiwa, Chiba, Japan) (AHs '22). ACM, New York, NY, USA, 142–151. <https://doi.org/10.1145/3519391.3519402>
- Koichi Shimizu, Gaku Sueta, Kentaro Yamaoka, Kazuki Sawamura, Yujin Suzuki, Keisuke Yoshida, Vibol Yem, Yasushi Ikei, Tomohiro Amemiya, Makoto Sato, Koichi Hirota, and Michiteru Kitazaki. 2018. FiveStar VR: shareable travel experience through multisensory stimulation to the whole body. In *SIGGRAPH Asia 2018 Virtual & Augmented Reality* (Tokyo, Japan) (SA '18). ACM, New York, NY, USA, Article 2, 2 pages. <https://doi.org/10.1145/3275495.3275502>
- Erwin Wu, Florian Perteneder, Hideki Koike, and Takayuki Nozawa. 2019. How to VizSki: Visualizing Captured Skier Motion in a VR Ski Training Simulator. In *Proceedings of the 17th International Conference on Virtual-Reality Continuum and Its Applications in Industry* (Brisbane, QLD, Australia) (VRCAI '19). Association for Computing Machinery, New York, NY, USA, Article 5, 9 pages. <https://doi.org/10.1145/3359997.3365698>